

Software Test Plan

Fashion E-commerce Web Application

Prepared in accordance with IEEE Std 829 (Test Plan outline)

Document ID	QT-FEC-STP-001
Version	1.0 (Baseline draft for review)
Date	25 September 2026
Application Under Test	https://fashion-ecommerce-we-lunb.bolt.host
Basis Document	Fashion E-commerce Application – QA User Stories & End-to-End Test PRD (60 user stories, 10 modules)
Standard	IEEE Std 829 – Standard for Software Test Documentation (Test Plan)
Prepared by	Ram – Senior Test Manager, Quality Thought
Status	Draft – pending stakeholder approval

Revision History

Version	Date	Author	Description
1.0	25 September 2026	Ram (Sr. Test Manager)	Initial test plan derived from the QA User Stories PRD

Table of Contents

1. Test Plan Identifier	3
2. References	3
3. Introduction	3
3.1 Purpose	3
3.2 Objectives	3
3.3 Scope Summary	3
4. Test Items	4
5. Software Risk Issues	4
6. Features to be Tested	5
6.1 Non-functional coverage (PRD Section 4)	6
7. Features Not to be Tested	6
8. Approach	7
8.1 Overall strategy	7
8.2 Test levels and types	7
8.3 Test design rules	7
8.4 Automation approach	8
8.5 Test cycles	8
8.6 Defect management	8
9. Item Pass / Fail Criteria	8
9.1 Test case level	8
9.2 User story level	8
9.3 Release (exit) criteria	9
10. Suspension Criteria and Resumption Requirements	9
11. Test Deliverables	10
12. Remaining Test Tasks	10
13. Environmental Needs	11
13.1 Test environment	11
13.2 Browser and device matrix	11
13.3 Tools	11
14. Staffing and Training Needs	11
Training	12
15. Responsibilities	12
16. Schedule	12
17. Planning Risks and Contingencies	13
18. Approvals	13
19. Glossary	14
Appendix A. Requirements Traceability Matrix (seed)	15
Appendix B. Core End-to-End Scenarios	17

1. Test Plan Identifier

Identifier	QT-FEC-STP-001, version 1.0
Plan level	Master Test Plan (system, integration-at-UI, end-to-end and acceptance-support levels)
Test item release	Build/release currently deployed at https://fashion-ecommerce-we-lunb.bolt.host (release tag to be recorded at test-readiness review)
Owner	Ram, Senior Test Manager – Quality Thought
Change control	Changes after baseline are raised as plan change requests, reviewed by the Test Manager and approved by the signatories in Section 18. The version number increments on each approved change.

2. References

Ref	Document	Relevance
R1	Fashion E-commerce Application – QA User Stories & End-to-End Test PRD	Primary test basis: 60 user stories (US-01 to US-60), acceptance criteria and priorities
R2	IEEE Std 829 – Standard for Software Test Documentation	Structure and content of this test plan and downstream test deliverables
R3	OWASP Top 10 & OWASP Web Security Testing Guide	Security test approach for US-38, US-58 and authentication stories
R4	WCAG 2.1 Level AA (W3C)	Accessibility acceptance thresholds for US-57
R5	PCI DSS guidance on cardholder data handling	Card-data exposure checks (US-34, US-38, US-48)
R6	Payment gateway sandbox documentation (to be supplied by development)	Test cards/UPI/wallet simulators for US-34, US-37, US-42, US-44
R7	Project defect management and test management workflow (Jira / TestRail or equivalent)	Defect lifecycle and reporting

3. Introduction

3.1 Purpose

This plan defines the scope, approach, resources, schedule and risks for testing the Fashion E-commerce web application before handoff and release. It translates the 60 user stories in the QA PRD (R1) into a controlled test effort that covers the complete customer journey – from landing and registration, through discovery, cart and checkout, to post-order support.

3.2 Objectives

- Verify every acceptance criterion in US-01 to US-60 with at least one positive and one negative/edge-case test.
- Prove the core purchase funnel (register/login → search → PDP → cart → checkout → payment → confirmation) end to end on every supported browser and device class.
- Confirm that transaction-critical and security-critical behaviour (auth, cart maths, payment, duplicate-order prevention, data protection) is defect-free at Critical/High severity.
- Provide objective, traceable evidence (Story ID → test case → result → defect) to support the go/no-go decision.

3.3 Scope Summary

The test basis contains **60 user stories** across **10 functional modules** and **145 acceptance criteria**. Applying the PRD rule of at least one positive and one negative case per criterion gives a **minimum of 290 test cases**; the planned suite is expected to be 20–30% larger once boundary and combination cases are added.

Module	Stories	Critical	High	Medium	Low	Acceptance criteria
M01 – Account Registration & Authentication	7	1	3	3	0	20
M02 – Home Page, Navigation & Content	4	0	0	2	2	10
M03 – Product Discovery: Search, Browse & Filters	6	1	1	3	1	16
M04 – Product Detail Page (PDP)	7	1	2	2	2	17
M05 – Shopping Cart	7	2	3	1	1	16
M06 – Checkout & Payments	8	5	1	1	1	19
M07 – Orders, Tracking, Returns & Cancellations	6	0	3	2	1	13
M08 – Account Management	4	0	0	4	0	10
M09 – Notifications, Promotions & Content	4	0	0	2	2	8
M10 – Cross-Cutting: Performance, Security, Compatibility & Accessibility	7	1	1	4	1	16
Total	60	11	14	24	11	145

Note (from R1): the PRD was written from standard fashion e-commerce behaviour because the client-rendered site could not be enumerated automatically. A site walkthrough against the PRD is a mandatory entry task (Section 12); any deviations are recorded and the affected acceptance criteria are updated before execution, rather than the story being skipped.

4. Test Items

The following items are submitted for test. Exact build identifiers are recorded in the Test Item Transmittal Report for each drop.

Item	Description	Version / Source
Web storefront (desktop)	Customer-facing responsive web application at https://fashion-ecommerce-we-lunb.bolt.host	Build under test (TBC at readiness review)
Web storefront (mobile web)	Same application rendered on mobile and tablet browsers	Same build
Authentication services	Email/password login, social OAuth (Google/Facebook), password reset, session management	Same build + configured OAuth apps
Catalogue & search	Listings, search, autocomplete, filters, sort, pagination, PDP data and inventory	Same build + test catalogue data
Cart, pricing & promotions	Cart, tax/shipping/discount calculation, coupons, gift cards/store credit	Same build + pricing/tax rules config
Checkout & payment integration	Address, shipping method, payment (card, UPI, wallet, COD) via gateway sandbox	Gateway sandbox
Order management (customer-facing)	Order history, tracking, cancellation, returns/exchanges, refunds, invoices	Same build + courier/refund stubs
Notifications	Transactional emails/SMS, newsletter, back-in-stock alerts	Email/SMS sandbox or capture tool

5. Software Risk Issues

Product risks drive test depth and sequencing. Risk = likelihood × business impact; the PRD priority is used as the impact baseline.

ID	Risk area	Why it matters	Stories	Level	Mitigation in testing
PR-01	Payment processing & duplicate charges	Direct financial loss and customer trust	US-34, US-36, US-37	High	Sandbox tests for every method; double-click, refresh and network-retry tests; idempotency checks
PR-02	Price / tax / discount calculation errors	Over- or under-charging; revenue leakage	US-29, US-30, US-39, US-52	High	Oracle spreadsheet of expected totals; min. 5 varied cart combinations; boundary coupons
PR-03	Authentication & session weaknesses	Account takeover; privacy breach	US-02, US-04-06, US-46	High	Lockout/throttle, token expiry, back-button, re-auth tests; OWASP checks
PR-04	Injection, XSS, CSRF, data exposure	Security incident; regulatory exposure	US-12, US-21, US-38, US-48, US-58	High	DAST scan (ZAP/Burp) + manual payloads; console/network/storage inspection
PR-05	Inventory / variant mismatch	Selling unavailable stock; wrong item shipped	US-19, US-20, US-26	High	Variant-to-SKU data checks; last-unit race condition with two users
PR-06	Unconfirmed business rules (PRD assumptions)	Tests built on wrong expectations	All	Medium	Mandatory site walkthrough; deviation log signed off by Product Owner
PR-07	Third-party dependencies (OAuth, email, courier, gateway)	Flows blocked by external systems	US-03, US-04, US-36, US-41, US-50	Medium	Sandboxes/stubs; early connectivity smoke test
PR-08	Responsive, browser and accessibility defects	Lost mobile conversions; exclusion of users	US-54, US-56, US-57	Medium	Device/browser matrix; real-device purchase; axe + screen-reader passes
PR-09	Performance of listing, PDP and checkout	Abandonment	US-13, US-16, US-55	Medium	Lighthouse baselines; concurrent-user spot-check on PDP/checkout

6. Features to be Tested

All 60 user stories are in scope. The table below groups them by module; the full story-level traceability matrix is in Appendix A.

Module	User stories in scope	Key test focus
M01 Account Registration & Authentication	US-01 – US-07 (7)	Field validation, password rules, duplicate email, lockout/throttling, OAuth, reset-link expiry, logout & back-button, session timeout, guest checkout & lookup
M02 Home Page, Navigation & Content	US-08 – US-11 (4)	Homepage sections and carousel, CTA destinations, nested navigation, keyboard/mobile menu, breadcrumbs, footer links, empty category states
M03 Product Discovery: Search, Browse & Filters	US-12 – US-17 (6)	Exact/partial/typo search, empty state, injection safety, autocomplete latency, single & combined filters and counts, sort persistence, pagination integrity, view toggle
M04 Product Detail Page (PDP)	US-18 – US-24 (7)	Data vs backend, gallery/zoom, size chart, variant → price/image/stock, out-of-stock handling, reviews rules, recommendations, wishlist toggle

Module	User stories in scope	Key test focus
M05 Shopping Cart	US-25 – US-31 (7)	Add/increment, quantity bounds vs stock, remove & empty state, persistence, subtotal/tax/shipping/discount maths, coupon rules, delivery estimate
M06 Checkout & Payments	US-32 – US-39 (8)	Address validation & saved addresses, shipping methods, all payment methods, card validation, failure & retry, review/edit, confirmation & email, duplicate prevention, HTTPS & data exposure, gift cards
M07 Orders, Tracking, Returns & Cancellations	US-40 – US-45 (6)	Order history & details, tracking, cancellation window & refund, return window/reasons/uploads, refund amount & status, invoice accuracy & PDF
M08 Account Management	US-46 – US-49 (4)	Profile updates with re-auth, address book & default, masked saved cards, wishlist persistence & move-to-cart
M09 Notifications, Promotions & Content	US-50 – US-53 (4)	Lifecycle notifications & content, newsletter subscribe/unsubscribe, sale price maths & badges, back-in-stock per variant
M10 Cross-Cutting: Performance, Security, Compatibility & Accessibility	US-54 – US-60 (7)	Responsive breakpoints, page performance, cross-browser (latest & latest-1), WCAG 2.1 AA, XSS/SQLi/CSRF/token storage, offline/timeout recovery, 404/500 pages

6.1 Non-functional coverage (PRD Section 4)

- Full purchase-funnel regression after every deployment.
- Tax/shipping/discount maths verified against manually calculated totals for at least five varied carts.
- Negative testing: invalid coupons, expired sessions, and the last-unit race condition (two users buying the final item simultaneously).
- At least one complete purchase on a real or emulated mobile device – not only a resized desktop browser.
- Localisation and currency formatting, if multiple regions are supported.
- Load/performance spot-check of PDP and checkout under concurrent simulated users.

7. Features Not to be Tested

Feature / area	Reason	Residual risk handling
Back-office admin, CMS and merchandising tools	Not part of the PRD test basis; customer storefront only	Test data set up via admin by development/data team
Internal logic of payment gateway, OAuth providers, courier and SMS/email vendors	Owned and certified by third parties	Integration verified through sandboxes and contract behaviour only
Real-money transactions and live courier shipments	Unsafe and irreversible in test	Sandbox payments, stubbed tracking/refund events; production smoke by business after go-live
Native iOS / Android apps	Not in the application under test	Mobile web covered under US-54
Full-scale load, stress and soak testing	PRD asks only for a spot-check; no performance environment confirmed	Spot-check in this plan; recommend a separate performance test plan before peak sales
Formal penetration test / certification (e.g. PCI audit)	Requires specialist, independently contracted testers	OWASP-based functional security testing in scope; formal pen test recommended pre-launch
Multi-region localisation	Conditional – only if the store supports multiple regions	Moved in scope if confirmed during site walkthrough

8. Approach

8.1 Overall strategy

A risk-based, requirements-driven approach is used. Test design is derived directly from the PRD acceptance criteria, executed in priority order (Critical → High → Medium → Low), with every story receiving at least one full pass before sign-off. Scripted testing is complemented by time-boxed exploratory sessions on high-risk modules.

8.2 Test levels and types

Code	Test type	Approach	Primary stories
SMK	Smoke / build verification	30-minute check of site availability, login, search, add-to-cart and sandbox checkout on each new build	Funnel stories
FUNC	Functional (system)	Scripted positive, negative and boundary tests per acceptance criterion using equivalence partitioning, boundary value analysis and decision tables	All
NEG	Negative / edge	Invalid input, expired links and sessions, stock limits, invalid coupons, failed payments	US-01-06, 12, 20, 26, 30, 34, 42, 43, 59, 60
DATA	Data & calculation	Independent oracle spreadsheet for totals, discounts, refunds, filter counts and sort order	US-14-16, 18-20, 29-31, 33, 39, 44, 45, 52
E2E	End-to-end journey	Registered, guest and social-login purchase journeys through to order, cancel, return and refund	US-07, 35, 43 + funnel
INT	Integration (at UI)	OAuth, email, SMS, gateway, courier and refund sandboxes/stubs	US-03, 04, 34, 36, 41, 42, 50, 51, 53
SEC	Security	OWASP-guided manual tests plus automated DAST scan; HTTPS/TLS, injection, XSS, CSRF, token storage, masking	US-02, 04-06, 12, 21, 38, 46, 48, 58
CONC	Concurrency	Parallel sessions for double-submit and last-unit race conditions	US-20, 37
XB / MOB	Cross-browser & responsive	Matrix in Section 13.2; all Critical stories re-verified on every browser/device class	US-54, 56 + all Critical
A11Y	Accessibility	Automated axe scan plus manual keyboard and screen-reader (NVDA / VoiceOver) passes against WCAG 2.1 AA	US-09, 57
PERF	Performance spot-check	Lighthouse/WebPageTest page metrics; k6/JMeter concurrent-user spot-check on PDP and checkout	US-13, 16, 55
RES	Resilience	Network throttling/offline simulation in DevTools during checkout and cart	US-59
REG	Regression	Automated funnel suite (Playwright/Selenium) on every deploy; full manual regression before release	Funnel + fixed defects

8.3 Test design rules

- Each acceptance criterion maps to ≥ 1 positive and ≥ 1 negative/edge test case; test case IDs follow **TC-US<nn>-<seq>** (e.g. TC-US19-03).
- Every test case and defect references its User Story ID for traceability.
- Where the live site differs from a PRD assumption, the tester logs a deviation (DEV-<nn>), the acceptance criterion is updated with Product Owner agreement, and the story is still tested.
- Critical stories (auth, cart, payment, security) are executed on every supported browser/device, not once.

8.4 Automation approach

The purchase funnel, login, search, cart calculation and checkout-validation tests are automated first, because they run on every deploy. Automation uses Playwright (or Selenium, per team skill) with a page-object model, runs against the test environment in CI, and publishes results to the test management tool. Target: $\geq 70\%$ of Critical and High regression cases automated by the end of cycle 2.

8.5 Test cycles

Cycle	Content	Exit to next cycle
Cycle 0 – Readiness	Environment, data and sandbox checks; site walkthrough vs PRD; smoke test	Smoke passes; deviation log agreed
Cycle 1 – Functional	All 11 Critical and 14 High stories, then Medium and Low; security and data tests start	100% planned tests executed once
Cycle 2 – Fix & cross-platform	Defect re-tests; cross-browser, mobile, accessibility, performance and concurrency	No open Critical; High agreed
Cycle 3 – Regression & UAT support	Full regression, E2E journeys, UAT support, release readiness	Exit criteria in Section 9 met

8.6 Defect management

Severity	Definition	Target fix / re-test
S1 – Critical	Blocks purchase, loses or corrupts data, charges incorrectly, or exposes security/personal data; no workaround	Fix within 24 h; re-test same day
S2 – High	Major feature broken or wrong result; workaround exists but is unacceptable to customers	Fix within 3 working days
S3 – Medium	Feature partly works; degraded experience with reasonable workaround	Fix within the current cycle
S4 – Low	Cosmetic or minor usability issue	Planned by Product Owner

Lifecycle: New → Triaged → Assigned → Fixed → Ready for Re-test → Closed / Reopened (plus Deferred / Rejected with Product Owner approval). Daily triage is held during execution. Each defect records Story ID, environment, browser/device, build, steps, expected vs actual, and evidence (screenshot, video or HAR file).

9. Item Pass / Fail Criteria

9.1 Test case level

A test case passes when the actual result matches the expected result derived from the acceptance criterion on the stated environment. Any deviation is a fail and is logged as a defect or a documented PRD deviation.

9.2 User story level

- **Pass:** all linked test cases pass, or remaining failures are S4 only and accepted by the Product Owner.
- **Critical stories:** pass on every browser/device in the Section 13.2 matrix.

9.3 Release (exit) criteria

#	Criterion	Target
1	Planned test cases executed	100%
2	Pass rate – Critical & High stories	100% of stories passed
3	Pass rate – overall test cases	≥ 95%
4	Open S1 (Critical) defects	0
5	Open S2 (High) defects	0, or ≤ 3 with approved workaround and fix date
6	Security findings (High/Critical per OWASP rating)	0 open
7	Cart / order total accuracy on oracle carts	100% match (≥ 5 cart combinations)
8	Duplicate order / charge tests	0 duplicates
9	Accessibility	No WCAG 2.1 AA failures on home, PLP, PDP, cart and checkout
10	Performance spot-check	Home, listing and PDP LCP ≤ 2.5 s; checkout step response ≤ 3 s (thresholds to be confirmed by Product Owner)
11	Traceability	All 60 stories have executed tests and a recorded result

10. Suspension Criteria and Resumption Requirements

Suspend testing when...	Resume when...
Smoke test fails on a new build (site down, login or add-to-cart broken)	A corrected build passes smoke test
An S1 defect blocks ≥ 30% of the remaining planned tests in a module	Fix deployed and verified; blocked tests re-planned
Test environment, payment sandbox or email capture unavailable for more than 4 working hours	Environment restored and a connectivity check passes
Test data is corrupted or inventory/pricing configuration differs from agreed baseline	Data restored to baseline and verified
Uncontrolled deployments occur during a test cycle	Build freeze re-established; impacted tests re-run
More than 25% of executed tests fail in the first day of a cycle (build not test-ready)	Development confirms fixes and a new build passes smoke test

11. Test Deliverables

Deliverable (IEEE 829)	Description	Owner	When
Test Plan	This document	Test Manager	Before test design
Test Design Specification	Test conditions and techniques per module	Test Lead	Cycle 0
Test Case Specification	≥ 290 test cases with Story ID traceability	Test Engineers	Cycle 0
Test Procedure Specification	E2E journey scripts and automation suite	Test Engineers / Automation	Cycle 0-1
Test Item Transmittal Report	Build ID, change notes and known issues per drop	Development Lead	Each build
Test Log	Chronological execution record with evidence	Test Engineers	Daily
Test Incident (Defect) Reports	Defects logged against User Story IDs	Test Engineers	Continuous
PRD Deviation Log	Live-site differences and updated acceptance criteria	Test Lead / Product Owner	Cycle 0, then as found
Daily / Weekly Status Report	Progress, pass rate, defect trends, risks	Test Manager	Daily / weekly
Requirements Traceability Matrix	Story → test case → result → defect (Appendix A seed)	Test Lead	Maintained throughout
Test Summary Report	Results versus exit criteria with go/no-go recommendation	Test Manager	End of Cycle 3

12. Remaining Test Tasks

Tasks that must be completed before or alongside execution:

#	Task	Owner	Dependency
1	Walk the live site against the PRD and publish the deviation log	Test Lead	Environment access
2	Confirm business rules: password policy, lockout threshold, session timeout, return window, cancellation statuses, coupon rules, tax and shipping tables	Product Owner	—
3	Confirm supported payment methods and obtain gateway sandbox credentials and test cards/UPI IDs	Development Lead	Gateway account
4	Set up OAuth test apps (Google/Facebook) and email/SMS capture	DevOps	—
5	Seed test data: catalogue with variants, low/zero-stock SKUs, sale items, coupons, gift cards, historical orders	Data / Dev team	Admin access
6	Confirm performance thresholds and supported browser/device list	Product Owner	—
7	Build oracle spreadsheet for totals, discounts and refunds	Test Engineer	Item 2
8	Set up automation framework, CI job and test management project	Automation Engineer	Tool licences

13. Environmental Needs

13.1 Test environment

- Application under test: <https://fashion-ecommerce-we-lunb.bolt.host> (or a dedicated staging copy with production-like configuration). Production must not be used for destructive tests.
- HTTPS with a valid certificate on all pages (required to test US-38).
- Payment gateway in sandbox mode supporting card, UPI, wallet and COD; courier tracking and refund events stubbed or sandboxed.
- Email and SMS capture (e.g. Mailtrap/MailHog or vendor sandbox) for registration, reset, confirmation, notification and newsletter tests.
- Ability to reset test data and adjust inventory/time (for session timeout, reset-link expiry and return windows).

13.2 Browser and device matrix

Platform	Browsers / devices	Coverage
Desktop – Windows 11	Chrome, Edge, Firefox (latest and latest-1)	Full suite on Chrome; Critical + smoke on others
Desktop – macOS	Safari, Chrome (latest and latest-1)	Critical + funnel
Mobile – Android	Chrome on a mid-range real device + emulators (360–412 px)	Funnel, responsive, touch; one full real-device purchase
Mobile – iOS	Safari on iPhone (real device or BrowserStack)	Funnel, responsive, touch
Tablet	iPad Safari, Android tablet Chrome (768–1024 px)	Responsive layout, navigation, checkout
Assistive tech	NVDA + Firefox/Chrome; VoiceOver + Safari; keyboard only	US-57 and key pages

13.3 Tools

Purpose	Tool (or approved equivalent)
Test & defect management	Jira + TestRail / Zephyr
UI automation	Playwright or Selenium WebDriver; CI via GitHub Actions / Jenkins
API / network inspection	Postman; browser DevTools; HAR capture
Security	OWASP ZAP, Burp Suite Community; browser storage inspection
Accessibility	axe DevTools, WAVE, Colour Contrast Analyser, NVDA, VoiceOver
Performance	Lighthouse, WebPageTest, k6 or JMeter
Cross-browser / devices	BrowserStack or LambdaTest, plus real devices
Network simulation	DevTools throttling/offline; proxy tools

14. Staffing and Training Needs

Role	Count	Key skills
Senior Test Manager	1	Planning, risk, reporting, stakeholder management
Test Lead	1	Test design, triage, traceability, e-commerce domain
Functional Test Engineers	3	Manual and exploratory testing, e-commerce flows, data validation
Automation Engineer	1	Playwright/Selenium, CI, page-object design
Security & Non-functional Tester	1 (part-time)	OWASP testing, ZAP/Burp, accessibility, performance tools

Training

- PRD walkthrough and business-rule briefing by the Product Owner (half day).
- Payment sandbox, test cards and refund flows briefing by the Development Lead.
- OWASP Top 10 and WCAG 2.1 AA refresher for testers new to security and accessibility testing.
- Automation framework onboarding for functional testers who will maintain scripts.

15. Responsibilities

Activity	Test Mgr	Test Lead	Testers	Automation	Dev Lead	Product Owner	DevOps
Approve test plan	R	C	I	I	C	A	I
Clarify requirements / deviations	I	R	C	I	C	A	-
Test design & test cases	A	R	R	C	I	C	-
Environment & sandboxes	A	C	I	C	R	I	R
Test data	A	R	C	I	R	C	C
Test execution	A	R	R	R	I	I	-
Defect triage & fixing	C	R	C	I	R	A	-
Status reporting	R/A	R	C	C	I	I	I
Go / no-go decision	R	C	I	I	C	A	I

R = Responsible, A = Accountable, C = Consulted, I = Informed

16. Schedule

Indicative six-week schedule, assuming test-ready build and environment on the start date. Dates are re-baselined at the test-readiness review.

Phase / milestone	Duration	Planned dates	Deliverable / exit
Test planning & plan approval	3 days	28 Sep – 30 Sep 2026	Approved test plan
Cycle 0 – Readiness, walkthrough, test design	5 days	01 Oct – 07 Oct 2026	≥ 290 test cases, deviation log, smoke pass
Cycle 1 – Functional execution (priority order)	8 days	08 Oct – 19 Oct 2026	All stories executed once
Cycle 2 – Fix verification & cross-platform / NFR	7 days	20 Oct – 28 Oct 2026	No open S1
Cycle 3 – Regression, E2E & UAT support	5 days	29 Oct – 04 Nov 2026	Exit criteria met
Test summary report & go / no-go	2 days	05 Nov – 06 Nov 2026	Signed Test Summary Report

17. Planning Risks and Contingencies

ID	Planning risk	Likelihood / impact	Contingency
RK-01	Live site differs significantly from PRD assumptions	High / High	Front-load walkthrough; update criteria with PO; re-scope estimates
RK-02	Late or unstable builds compress the schedule	Medium / High	Priority-order execution so Critical/High finish first; add overtime or temporary testers
RK-03	Payment, OAuth or email sandboxes unavailable	Medium / High	Request early; use stubs; move dependent tests later in cycle
RK-04	Business rules (tax, returns, timeouts) not confirmed	Medium / Medium	Log assumptions; test against documented assumption; flag in status report
RK-05	Insufficient real devices / browser licences	Low / Medium	Cloud device lab (BrowserStack/LambdaTest)
RK-06	Key tester unavailability	Low / Medium	Cross-train across modules; maintain up-to-date test cases
RK-07	High defect volume exceeds fix capacity	Medium / High	Daily triage; PO agrees deferrals; extend Cycle 2
RK-08	Scope creep (new stories mid-cycle)	Medium / Medium	Change control via Section 1; impact assessed before acceptance

18. Approvals

Signature below indicates agreement with the scope, approach, schedule and exit criteria of this test plan.

Role	Name	Signature	Date
Senior Test Manager (Author)	Ram		
Product Owner			
Development / Engineering Lead			
Project / Delivery Manager			
Client Representative			

19. Glossary

Term	Definition
AUT	Application Under Test
PRD	Product Requirements Document – here, the QA User Stories & E2E Test PRD
PDP / PLP	Product Detail Page / Product Listing Page
E2E	End-to-end – a complete customer journey across modules
COD / UPI	Cash on Delivery / Unified Payments Interface
OAuth	Open standard used for social login (Google/Facebook)
XSS / SQLi / CSRF	Cross-Site Scripting / SQL Injection / Cross-Site Request Forgery
OWASP	Open Worldwide Application Security Project
WCAG 2.1 AA	Web Content Accessibility Guidelines, conformance level AA
LCP	Largest Contentful Paint – page load performance metric
DAST	Dynamic Application Security Testing
Oracle	Independent source of expected results (e.g. calculation spreadsheet)
RTM	Requirements Traceability Matrix
S1-S4	Defect severity levels defined in Section 8.6
Latest-1	The current and immediately previous major browser version

Appendix A. Requirements Traceability Matrix (seed)

One row per user story. **Min. TCs** = acceptance criteria × 2 (positive + negative). **XB** column marks stories that must be re-verified on every browser/device. Test case IDs, results and defect IDs are added during execution.

Type codes: *FUNC* functional · *NEG* negative · *DATA* calculation/data · *E2E* end-to-end · *INT* integration · *SEC* security · *CONC* concurrency · *SESS* session/persistence · *XB* cross-browser · *MOB* mobile · *UI* layout · *A11Y* accessibility · *PERF* performance · *RES* resilience

ID	User story (short)	Mod.	Priority	AC	Min. TCs	Test types	Re-verify all browsers
US-01	Register account with email & password	M01	High	4	8	FUNC, NEG, UI	–
US-02	Login with email & password	M01	Critical	3	6	FUNC, NEG, SEC, XB	Yes
US-03	Social / third-party login (OAuth)	M01	Medium	3	6	FUNC, INT	–
US-04	Forgot / reset password via email	M01	High	3	6	FUNC, NEG, SEC, INT	–
US-05	Logout and session clearing	M01	Medium	2	4	FUNC, SEC	–
US-06	Session timeout & re-authentication	M01	Medium	2	4	FUNC, SEC	–
US-07	Guest checkout	M01	High	3	6	FUNC, E2E	–
US-08	Homepage banners, collections, categories	M02	Medium	3	6	FUNC, UI	–
US-09	Main navigation menu	M02	Medium	3	6	FUNC, UI, A11Y	–
US-10	Breadcrumb navigation	M02	Low	2	4	FUNC, UI	–
US-11	Footer policy links	M02	Low	2	4	FUNC, UI	–
US-12	Keyword search	M03	Critical	4	8	FUNC, NEG, SEC, XB	Yes
US-13	Autocomplete / type-ahead	M03	Medium	2	4	FUNC, PERF	–
US-14	Filter listings (size, colour, price, brand, category)	M03	High	4	8	FUNC, DATA	–
US-15	Sort listings	M03	Medium	2	4	FUNC, DATA	–
US-16	Pagination / infinite scroll	M03	Medium	3	6	FUNC, PERF	–
US-17	Grid / list view toggle	M03	Low	1	2	UI	–
US-18	Detailed product information	M04	High	3	6	FUNC, DATA, UI	–
US-19	Size & colour variant selection	M04	Critical	3	6	FUNC, DATA, XB	Yes
US-20	Real-time stock availability	M04	High	2	4	FUNC, DATA, NEG	–
US-21	Ratings & reviews	M04	Medium	3	6	FUNC, SEC	–
US-22	Related / recommended products	M04	Low	2	4	FUNC, UI	–
US-23	Image zoom / 360° / multi-angle	M04	Low	2	4	UI, MOB	–
US-24	Add to wishlist from PDP / listing	M04	Medium	2	4	FUNC	–
US-25	Add product to cart	M05	Critical	3	6	FUNC, DATA, XB	Yes
US-26	Update cart quantity	M05	High	3	6	FUNC, NEG, DATA	–
US-27	Remove item from cart	M05	High	2	4	FUNC	–

ID	User story (short)	Mod.	Priority	AC	Min. TCs	Test types	Re-verify all browsers
US-28	Cart persistence across sessions / devices	M05	Medium	2	4	FUNC, SESS	–
US-29	Order summary calculations	M05	Critical	2	4	DATA, FUNC, XB	Yes
US-30	Promo / coupon code	M05	High	3	6	FUNC, NEG, DATA	–
US-31	Estimated delivery date / shipping cost	M05	Low	1	2	FUNC, DATA	–
US-32	Shipping address entry / selection	M06	Critical	3	6	FUNC, NEG, XB	Yes
US-33	Shipping method selection	M06	Medium	2	4	FUNC, DATA	–
US-34	Payment details (card, UPI, wallet, COD)	M06	Critical	3	6	FUNC, NEG, INT, SEC, XB	Yes
US-35	Order review before submission	M06	High	2	4	FUNC, E2E	–
US-36	Order confirmation (screen & email)	M06	Critical	3	6	FUNC, INT, NEG, XB	Yes
US-37	Duplicate-charge prevention	M06	Critical	2	4	NEG, CONC, XB	Yes
US-38	Secure transmission of payment / personal data	M06	Critical	2	4	SEC, XB	Yes
US-39	Gift card / store credit	M06	Low	2	4	FUNC, DATA	–
US-40	Order history	M07	High	2	4	FUNC, DATA	–
US-41	Shipment tracking	M07	Medium	2	4	FUNC, INT	–
US-42	Cancel order before shipping	M07	High	2	4	FUNC, NEG, INT	–
US-43	Return / exchange request	M07	High	3	6	FUNC, NEG, E2E	–
US-44	Refund status and amount	M07	Medium	2	4	FUNC, DATA	–
US-45	Invoice view / download	M07	Low	2	4	FUNC, DATA	–
US-46	Update profile details	M08	Medium	3	6	FUNC, NEG, SEC	–
US-47	Manage saved addresses	M08	Medium	2	4	FUNC	–
US-48	Save / manage payment methods	M08	Medium	2	4	FUNC, SEC	–
US-49	View / manage wishlist	M08	Medium	3	6	FUNC, SESS	–
US-50	Order status notifications (email/SMS)	M09	Medium	2	4	INT, DATA	–
US-51	Newsletter subscribe / unsubscribe	M09	Low	2	4	FUNC, NEG, INT	–
US-52	Sale / discount price display	M09	Medium	2	4	DATA, UI	–
US-53	Back-in-stock notification	M09	Low	2	4	FUNC, INT	–
US-54	Responsive design (mobile/tablet/desktop)	M10	High	2	4	MOB, UI	–
US-55	Page load performance	M10	Medium	2	4	PERF	–
US-56	Cross-browser compatibility	M10	Medium	2	4	XB	–
US-57	Accessibility (keyboard, screen reader, WCAG AA)	M10	Medium	3	6	A11Y	–
US-58	Protection against XSS, SQLi, CSRF	M10	Critical	3	6	SEC	Yes

ID	User story (short)	Mod.	Priority	AC	Min. TCs	Test types	Re-verify all browsers
US-59	Network interruption / timeout handling	M10	Medium	2	4	NEG, RES	–
US-60	Friendly 404 / 500 error pages	M10	Low	2	4	NEG, UI	–
	Total			145	290		11

Appendix B. Core End-to-End Scenarios

E2E ID	Scenario	Stories covered
E2E-01	New user registers, searches, filters, selects variant, adds to cart, applies coupon, pays by card, receives confirmation email	US-01, 12, 14, 19, 25, 29, 30, 32, 34–36
E2E-02	Guest shopper checks out with UPI, then retrieves order via order number + email and creates an account	US-07, 25, 32–36
E2E-03	Returning user logs in via Google, moves wishlist item to cart, uses saved address and card, pays with gift card + card split	US-03, 24, 47–49, 39, 34
E2E-04	Registered user cancels an unshipped order and verifies refund status and notification	US-40, 42, 44, 50
E2E-05	Delivered order returned within window; pickup, receipt and refund tracked; invoice downloaded	US-40, 43–45, 50
E2E-06	Payment fails then retried successfully without losing the cart; network drops during checkout and recovers	US-34, 37, 59
E2E-07	Two users attempt to buy the last unit of a size simultaneously	US-20, 26, 37
E2E-08	Full funnel on real mobile device and with keyboard/screen reader only	US-54, 57 + funnel